



Computer Engineering Department

Course name: Web Technologies

Course code: 2CP08

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Topic: Report of LAB - 5

LAB 3: Cascading Style Sheets (CSS) @import CSS, Apply style to ordered List, Descendant Selectors, child selectors, first-line, first-letter, :before, :after pseudo element, positioning, BOX Model

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5.1 Brief introduction of used CSS::

1. Positioning:

- **Static** (default): Elements appear in the normal document flow.
- **Relative**: Positioned relative to its normal position without affecting other elements.
- **Absolute**: Removed from the document flow and positioned relative to the nearest positioned ancestor (or <html> if no ancestor is positioned).
- **Fixed**: Stays fixed relative to the viewport, unaffected by scrolling.
- **Sticky**: Switches between relative and fixed depending on scroll position.

2. Z-index:

- Elements with a **higher** z-index appear above those with a lower value.
- Work only with Following positions: relative, absolute, fixed, sticky
- Can have **positive**, **negative**, or **zero** values

3. Descendant Selector:

- It is used to select all nested elements inside parent.
- Example: `div p { /*css*/ }`
- It selects all p tags inside div.

4. Child Selector:

- Selects only direct child elements of a parent.
- Example: `p > a { /*css*/ }`
- It selects only that a who is direct child of p.

5. Adjacent Sibling Selector:

- Selects an element that is immediately next to another element.
- Example: `h1 + p { /*css*/ }`
- first p immediately following an h1.

6. CSS Box Model:

- The CSS Box Model describes how elements are structured and spaced in a webpage.
- Content: The actual text, image, or other content inside the element.
- Padding: Space inside the element, between the content and the border.
- Border: A boundary around the element.
- Margin: Space outside the element, separating it from other elements

7. OL:

- `list-style-type`: used for changing style of list.
- `text-indent`: used in li for giving indentation.

5.2 Exercise 1:

HTML:

```
<HTML>

  <HEAD>

    <TITLE> OL STYLING </TITLE>

    <link rel="stylesheet" href="style.css">

  </HEAD>

  <BODY>

    <P> Title: Web Technologies </P>

    <P>Table of content </P>

    <OL>

      <LI> HTML </LI>

      <LI> CSS </LI>

      <OL>

        <LI> INTRODUCTION </LI>

        <LI> ADDING STYLESHEET </LI>

        <OL>

          <LI> EXTERNAL </LI>

          <OL>

            <LI> EXTERNAL STYLE SHEETS ARE USEFUL </LI>

          </OL>

        </OL>

      <LI> BROWSER INCOMPATIBILITY </LI>

    </OL>
```

```
        <LI> XML </LI>
    </OL>
</BODY>
</HTML>
```

CSS:

```
body{
    text-transform: lowercase;
}
```

```
ol{
    list-style-type : none;
}
```

```
body> ol{
    counter-reset : cnt1;
}
```

```
body> ol> ol{
    counter-reset : cnt2;
}
```

```
body> ol> ol> ol {
    counter-reset : cnt3;
}
```

```
body> ol> ol> ol> ol{
    counter-reset : cnt4;
}
```

```
body> ol>li::before{
    counter-increment : cnt1;
    content: "Chapter " counter(cnt1)".";
}
```

```
}
```

```
body> ol> ol>li::before{  
    counter-increment : cnt2;  
    content: counter(cnt1)".counter(cnt2);  
}
```

```
body> ol> ol> ol>li::before {  
    counter-increment : cnt3;  
    content: counter(cnt1)".counter(cnt2)".counter(cnt3);  
}
```

```
body> ol> ol> ol> ol>li::before {  
    counter-increment : cnt4;  
    content: counter(cnt1)".counter(cnt2)".counter(cnt3)".counter(cnt4);  
}
```

OUTPUT:

title: web technologies

table of content

chapter 1. html

chapter 2. css

2.1 introduction

2.2 adding stylesheet

2.2.1 external

2.2.1.1 external style sheets are useful

2.3 browser incompatibility

chapter 3. xml

5.3 Exercise 2:

HTML:

```
<html>

<head>

  <title> layout </title>

  <link rel="stylesheet" href="style2.css">

</head>

<body>

  <div id="header"> <h1>Header</h1> </div>

  <div id="left"><h1>Left</h1> </div>

  <div id="main"><h1>Main</h1> </div>

</body>

</html>
```

CSS:

```
*{

  padding:0;

  margin:0;

}

#header{

  background : #f8f8f8;

  width: 100vw;

  height:20vh;

  position: sticky;

  top: 0;

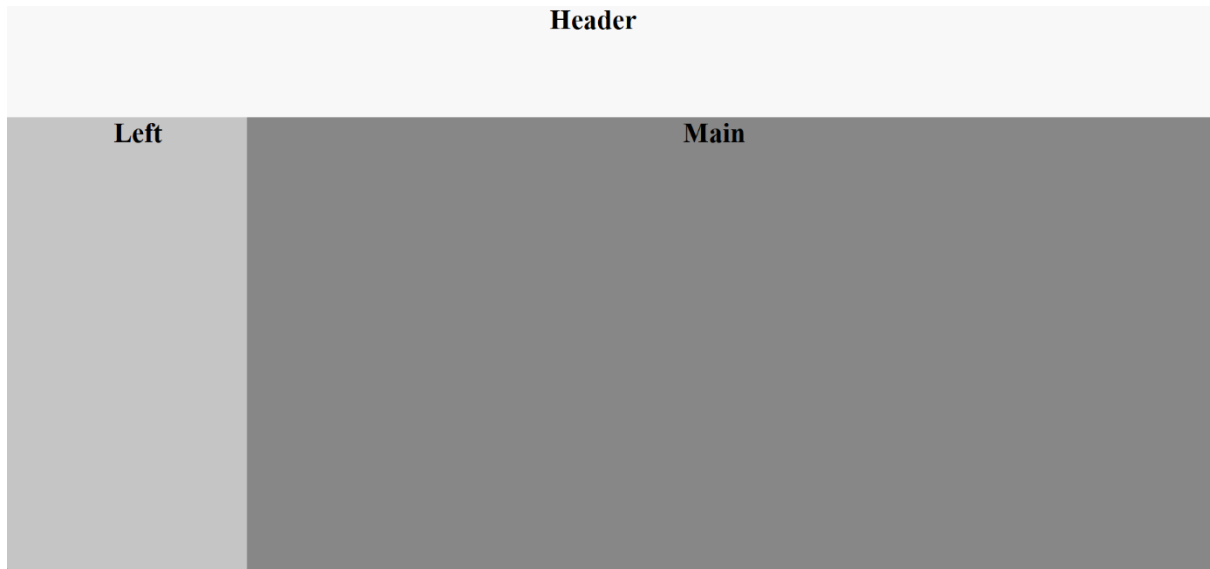
}
```

```
#left{
    height: 80vh;
    width: 20vw;
    display: inline-block;
    background: #c5c5c5;
    position: absolute;
    left: 0;
    bottom: 0;
}

#main{
    height: 80vh;
    width: 80vw;
    display: inline-block;
    background: #878787;
    position: absolute;
    right: 0;
    bottom: 0;
    z-index: 3;
}

h1{
    display: block;
    width: 10%;
    margin: auto;
}
```


OUTPUT:



5.4 Exercise 3:

HTML:

```
<html>
  <head>
    <title> Box model </title>
    <link rel="stylesheet" href="style3.css">
  </head>
  <body>
    <div>
      <h1> Local Finder </h1>
      <ul>
        <li> help user to check their needed product nearby </li>
        <li> gives experience of comparing prices at comfort of home </li>
        <li> physical shopping expirence </li>
      </ul>
    </div>
  </body>
</html>
```

CSS:

```
div{
  border: 3px solid #222;
  border-radius: 7px;
  padding : 20px;
  width: 500px;
  margin: auto;
  margin-top : 20vh;
  font-family: Arial;
}
div:hover{
  background : linear-gradient(#023992,#F5FFFA);
}
h1:hover{
  text-decoration : underline;
  font-family: 'Times New Roman';
}
li:hover{
  color : #010101;
  font-size: 110%;
  background : #CCDDDD;
}
```

OUTPUT:

Local Finder

- help user to check their needed product nearby
- gives experience of comparing prices at comfort of home
- physical shopping experience

Local Finder

- help user to check their needed product nearby
- gives experience of comparing prices at comfort of home
- physical shopping experience